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Sec: 06-A

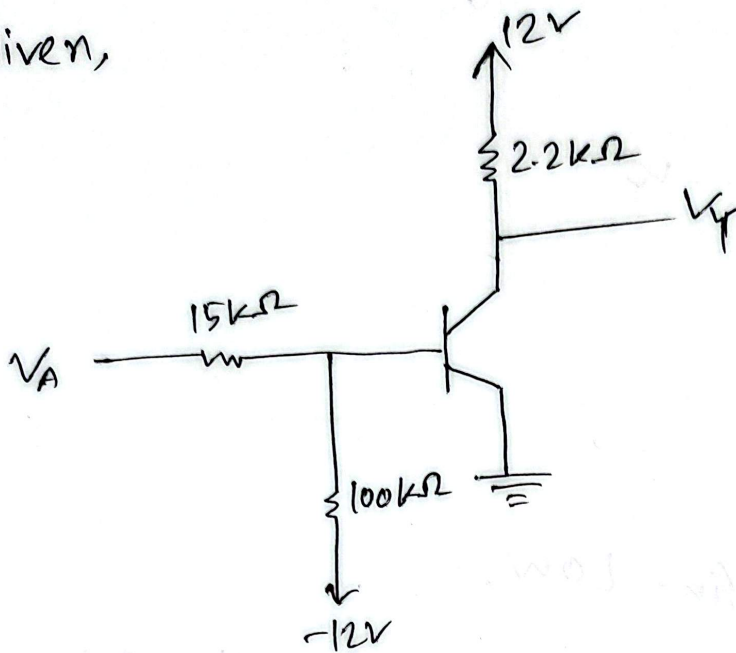
Course: CSE 350

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Problem-4

Given,



Since, this is an RTL NOT gate, the BJT is in cutoff mode for $V_{in} = \text{LOW}$ and saturation for

$V_{in} = \text{HIGH}$. So,

Case-0, $V_{in} = \text{LOW}$,

$$P_0 = 0.2 - (-12) \times \frac{0.2 - (-12)}{15 + 100}$$

$$= 1.294 \text{ mW}$$

Case-1
 $V_{in} = \text{HIGH}$

$$P_1 = 12 - (-12) \times \frac{0.8 - (-12)}{100} + (12 - 0) \times \left(\frac{12 - 0.8}{15} - \frac{0.8 + 12}{100} \right)$$

$$+ 12 - 0 \times \frac{12 - 0.2}{2.2}$$

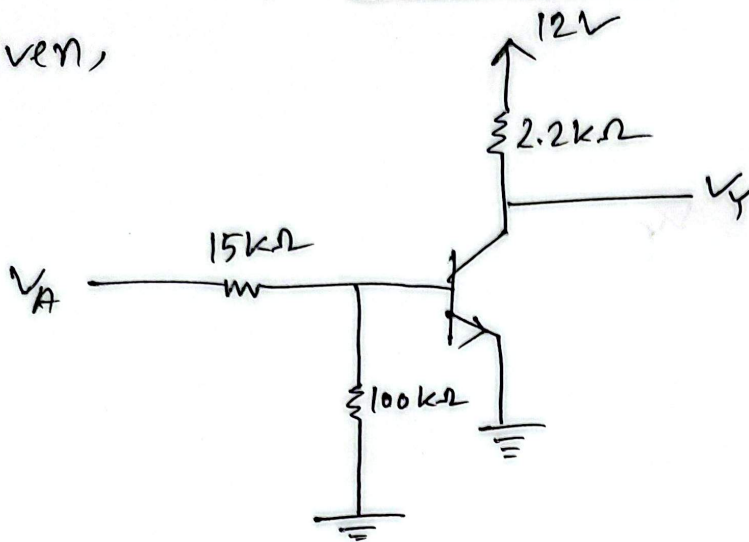
$$= 74.86 \text{ mW}$$

$$\therefore P_{\max} = 74.86 \text{ mW}$$

$$\therefore P_{\text{avg}} = \frac{74.86 + 1.294}{2} = 38.076 \text{ mW} \quad (\text{Ans.})$$

Problem-5

Given,



Similarly, for Case-0, $V_{in} = \text{LOW}$,

$$P_0 = \frac{0.2 - 0}{15 + 100} \times (0.2 - 0) = 3.478 \times 10^{-4} \text{ mW}$$

for Case-1, $V_{in} = \text{High}$,

$$P_L = (12 - 0) \times \left(\frac{0.8 - 0}{100} \right) + (12 - 0) \times \left(\frac{12 - 0.8}{15} - \frac{0.8 - 0}{100} \right) + (12 - 0) \times \left(\frac{12 - 0.2}{2.2} \right)$$
$$= 73.32 \text{ mW}$$

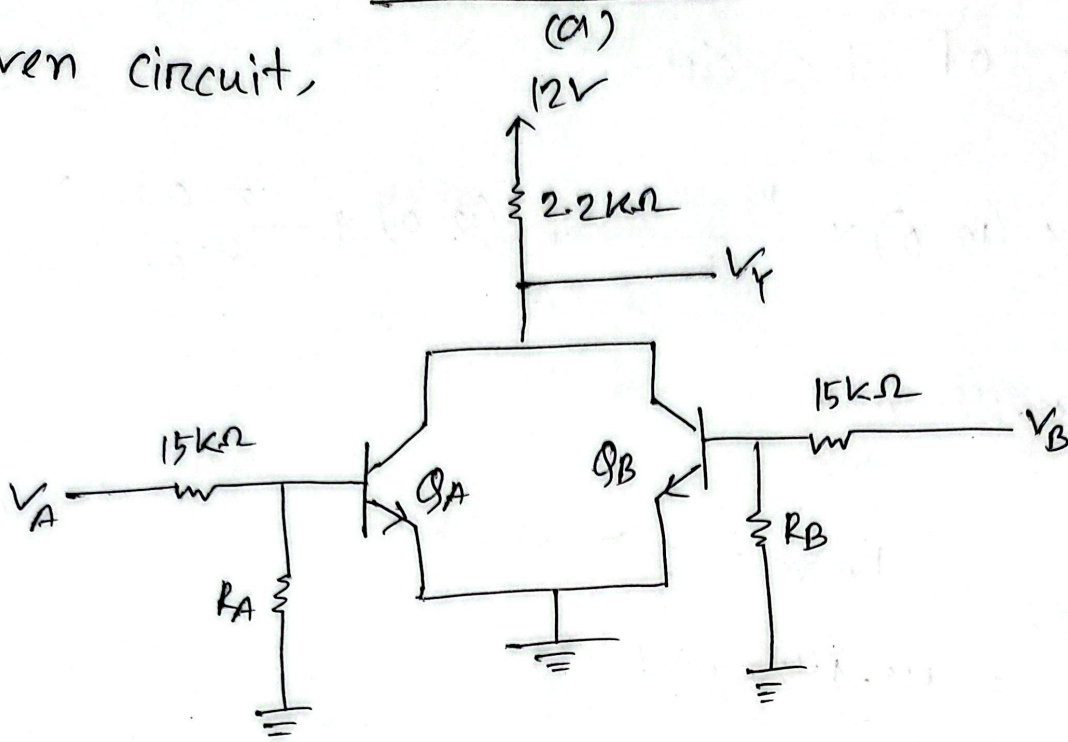
$$\therefore P_{\max} = 73.32 \text{ mW} \text{ and } P_{\min} = 3.478 \times 10^{-4} \text{ mW}$$

$$\text{the ratio, } \frac{P_{\max}}{P_{\min}} = \frac{73.32}{3.478 \times 10^{-4}} = 210805.45$$

(Ans.)

Problem-6

Given circuit,



For case (0,0), both Q_A and Q_B are in cutoff mode, since this is an RTL NOR Gate. So,

$$P_{(0,0)} = (0.2 - 0) \times \frac{0.2 - 0}{R_A + 15} + (0.2 - 0) \times \left(\frac{0.2 - 0}{R_B + 15} \right)$$

assuming that low-input = 0.2V and $R_A = R_B$ (given),

$$P_{(0,0)} = \frac{0.04 + 0.04}{R_A + 15}$$

$$\Rightarrow 0.5 \times 10^{-3} = \frac{0.08}{R_A + 15}$$

$$\Rightarrow R_A + 15 = \frac{0.08}{0.5 \times 10^{-3}} = 160$$

$$\therefore R_A = 145 \text{ k}\Omega$$

(b)

Maximum Power of the circuit,

$$P_{\max} = 2 \times (12-0) \times \frac{12-0.8}{15} + (12-0) \times \left(\frac{12-0.2}{2.2} \right)$$
$$= 82.284 \text{ mW}$$

Now,

$$P'_{\max} = \frac{1}{2} P_{\max} = \frac{1}{2} \times 82.284$$
$$= 41.142 \text{ mW}$$

$$\therefore P'_{\max} = 12 \times \left(\frac{12-0.8}{15} \right) \times 2 + (12-0) \times \frac{(12-0.2)}{R_c}$$

$$41.142 = 17.92 + \frac{141.6}{R_c}$$

$$\Rightarrow R_c = \frac{141.6}{41.142 - 17.92}$$
$$= 6.098 \text{ k}\Omega$$

$\therefore R_c$'s new value is $6.098 \text{ k}\Omega$.

(Ans.)